ROBOTICS & COMPUTER VISION LABORATORY

Vision-Based Autonomous Robotic Arm for Object Detection & Pick-and-Place

Integrating speech recognition, wrist-mounted OpenCV detection, and inverse kinematics control in CoppeliaSim.



PROJECT TEAM

[Student Names / Roll Nos]

Dept. of Electrical & Computer Engineering

PROJECT GUIDE

[Guide Name & Designation]

[College / University Name]

Project Abstract

Autonomous multimodal object selection bridging human speech and closed-loop robotic manipulation.

This project implements a complete simulation framework pairing speech processing with eye-in-hand computer vision to detect, localize, and sort colored cubes without manual coordinates or hardcoded waypoints.

Hands-Free Voice Interface

Eliminates teach pendants and fixed waypoint programming.

Eye-In-Hand Vision

Dynamic HSV thresholding adjusts for object positions in the workspace.

Kinematic Motion Planning

6-DOF analytical inverse kinematics calculates target joint angles.

Problem Statement

Traditional Limitations

- **Rigid Waypoints:** Hardcoded paths fail when target objects shift position.
- **Manual Control:** Operator joysticks and teach pendants are slow and prone to fatigue.
- **No Visual Feedback:** Conventional blind grippers cannot verify presence or color.

Proposed Solution

- **Natural Language:** Simple vocal instructions trigger automated picking.
- **Dynamic Detection:** Wrist-mounted camera tracks red, yellow, and blue cubes.
- **Adaptive IK:** Coordinates calculated dynamically for arbitrary workspace points.

Project Objectives

01

Voice Recognition

Parse spoken commands ("red", "yellow", "blue") into machine-readable color tokens.

02

Color Vision

Extract RGB frames, convert to HSV space, and filter target color masks via OpenCV.

03

Spatial Mapping

Compute 2D centroid coordinates and project to 3D world poses for manipulator reach.

04

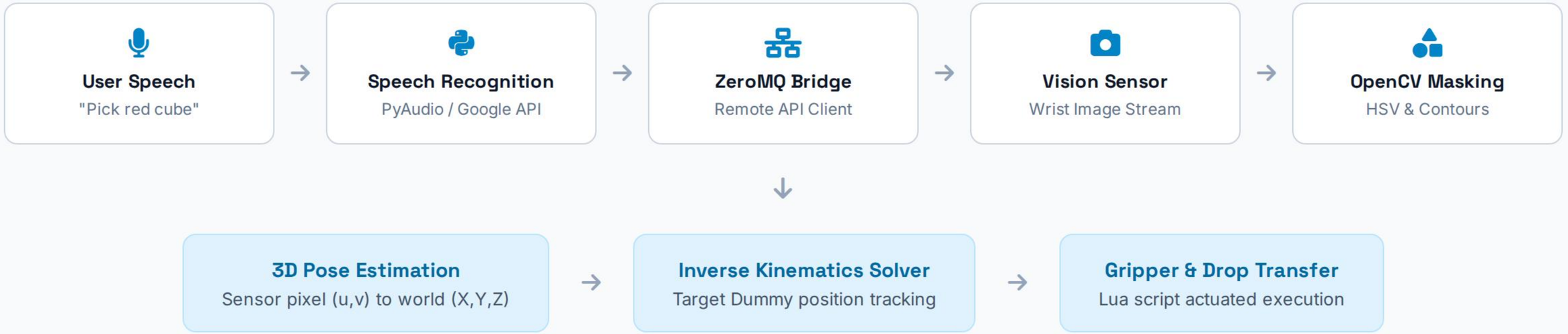
IK & Grasping

Execute inverse kinematics trajectory, actuate two-finger gripper, and transfer to drop box.

Deliverable: End-to-end simulated pick-and-place pipeline

Success metric: Correct color classification & target pose reach

System Architecture & Data Pipeline



Hardware & Software Environment

Robotic & Virtual Hardware

Manipulator	Niryo One (6-DOF open-source arm)
End-Effector	Standard Two-Finger parallel gripper
Vision Sensor	Wrist-mounted RGB (640×480 resolution)
Workpiece	Calibrated rigid cubes (Red, Yellow, Blue)

Software Framework

Simulator	CoppeliaSim Edu (v4.3+ engine)
Primary Code	Python 3.10 with SpeechRecognition library
Vision Library	OpenCV 4.x & NumPy matrix routines
Middleware	CoppeliaSim ZeroMQ Remote API + Lua scripts

Vision-Based Object Detection Pipeline

1. HSV Color Conversion

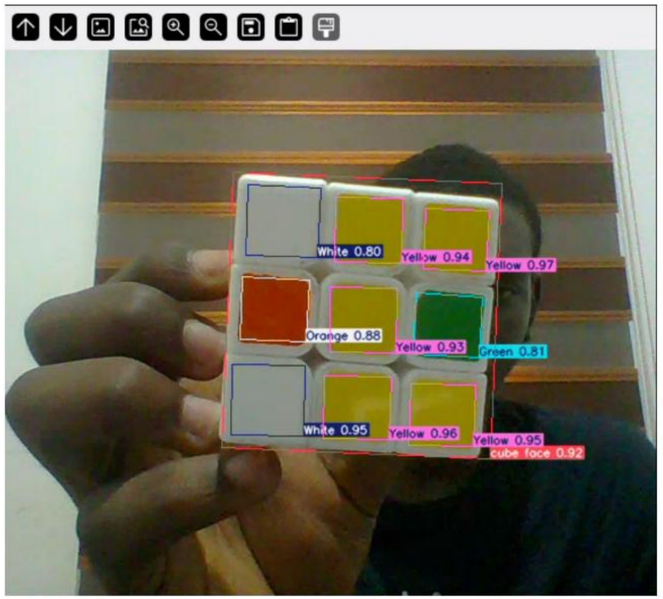
Transforms RGB raw frames to Hue-Saturation-Value to isolate chromatic data from ambient illumination changes.

2. Thresholding & Morphological Noise Filtering

Applies predefined lower/upper HSV bounds followed by opening and closing filters to remove pixel noise.

3. Contour Extraction & Bounding Boxes

Calculates area moments to determine centroid `(cx, cy)` and draws bounding rectangles around target cubes.



Voice Command Processing Pipeline

1. Audio Acquisition

Microphone captures ambient acoustic signal; ambient noise thresholding applied dynamically.

2. Speech-to-Text

Python `speech_recognition` library transcribes spoken audio into a string literal.

3. Intent Extraction

Keyword parsing matches tokens against defined vocabulary: `{"red", "yellow", "blue"}`.

4. API Dispatch

Sends active color code to CoppeliaSim via ZeroMQ to trigger specific color mask routing.

Example phrase: "Please pick up the yellow cube" → Action: Color Target = YELLOW

Fallback: Re-prompt on low confidence

Inverse Kinematics & Trajectory Planning

IK Tip & Target Mechanism

CoppeliaSim calculates 6-joint configurations using an IK group linking the gripper tip dummy to the target pose dummy.

Multi-Waypoint Trajectory

Generates an approach waypoint directly above the target (Z + 8cm) prior to lowering to avoid lateral workspace collisions.

Joint Limits & Damped Least Squares

Applies DLS solver methods to eliminate kinematic singularities and ensure all revolute joint positions remain within limits.



Kinematic Model Coordinates

$$T_{base}^{tool} = f(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)$$

Niryo One 6-DOF Revolute Articulation

Pick-and-Place Execution Sequence

01

Hover Approach

Move arm to standby position directly above detected cube.

02

Vertical Descent

Lower end-effector to object grasping height (Z offset).

03

Close Gripper

Actuate parallel fingers to clamp cube surfaces firmly.

04

Lift Object

Raise arm vertically to clear surrounding workspace items.

05

Transfer

Interpolate trajectory to the designated drop-off station.

06

Release

Open gripper fingers and return manipulator to home pose.

Implementation Status & Testing

Successfully Demonstrated

- **ZeroMQ API Bridge:** Stable bilateral communication established.
- **Voice Recognition:** Accurate keyword parsing for primary colors.
- **Vision Pipeline:** OpenCV HSV masking & 2D centroid extraction.
- **Arm Positioning:** Inverse kinematics reaches target coordinates.

Active Development & Limitations

- **Physics Tuning:** Gripper friction contact requires fine-tuning.
- **Depth Estimation:** Single monocular camera lacks true Z-depth sensing.
- **Lighting Sensitivity:** Extreme lighting changes require manual HSV tuning.
- **Real-World Deployment:** Transfer to physical Niryo One hardware pending.

[Screenshot: CoppeliaSim Scene] [Screenshot: OpenCV Mask Filter] [Screenshot: Terminal Voice Log] [Screenshot: IK Motion Path]

Summary & Future Scope

The system validates a modular architecture connecting voice intent, real-time image processing, and robotic arm inverse kinematics. It lays a direct foundation for agile industrial sorting and assistive manipulation.

Planned Engineering Enhancements

- RGB-D Depth Camera Integration
- YOLOv8 Dynamic Object Sorting
- Physical Hardware Deployment
- Obstacle Collision Avoidance



Thank You!

Open for Questions and Discussion

Image Sources



https://i.kickstarter.com/assets/015/696/879/b9078680270bc3d2c611506d4d034745_original.JPG?anim=false&fit=cover&gravity=auto&height=873&origin=ugc&q=92&v=1488373085&width=1552&sig=4JGrZZkehEejaD3xU2pGfTFD9fNkM4F%2BFqgDdTyi1lQ%3D

Source: www.kickstarter.com



https://miro.medium.com/v2/resize:fit:1400/1*8z8gY5_2_fBBcpo8AHWy7A.png

Source: medium.com